Lecture 4

NLP: Text Vectorization and Word Embeddings

Advanced Topics in Artificial Intelligence · UFABC

Adapted from MIT 15.773 (Farias, Ramakrishnan) — OCW

Part 1 — Why NLP?

Lecture roadmap

Part 1 — Why NLP? Motivation, applications and historical arc

Part 2 — Text Pre-processing Pipeline Standardize → Tokenize → Index → Encode

Part 3 — Bag of Words Count encoding, multi-hot, limitations

Part 4 — Word Embeddings One-hot problems, Word2Vec, GloVe

Part 5 — Embeddings in Code `nn.Embedding` (PyTorch), `Embedding` (Keras)

Part 6 — Application Text classification with embeddings

Why NLP?

Human knowledge lives in natural language

- The Internet is, for the most part, **text**
- Human communication and cultural production are **text**
- Corporate documents, contracts, medical records are **text**

Imagine a system that could automatically read and "understand" all of this — extracting patterns, classifying, answering questions, generating content.

Applications in production

🏷️ **Classification** Sentiment, intent, routing	📋 **Extraction** Financials, form data
📝 **Summarization** Abstracts, bullet points, titles	✍️ **Generation** Emails, reports, code
💬 **Q&A / RAG** Chatbots, semantic search	🌐 **Translation** Multilingual, code-switching

Historical arc of NLP

📜

Rules

up to 1990

- Formal grammars
- Manual syntax parsers
- Performance limited by rule scale

📊

Statistical / ML

1990 – 2013

- Bag of Words + SVM
- Hidden Markov Models
- Logistic regression over n-grams

🧠

Neural Networks

2014 – 2017

- RNNs and LSTMs
- Word2Vec, GloVe
- Seq2seq with attention

⚡

Transformers

2017 – today

- "Attention is All You Need"
- BERT, GPT, T5, LLaMA
- General-purpose LLMs

"Every time I fire a linguist, the performance of the speech recognizer goes up." — Frederick Jelinek, IBM

Big picture: NLP as regression

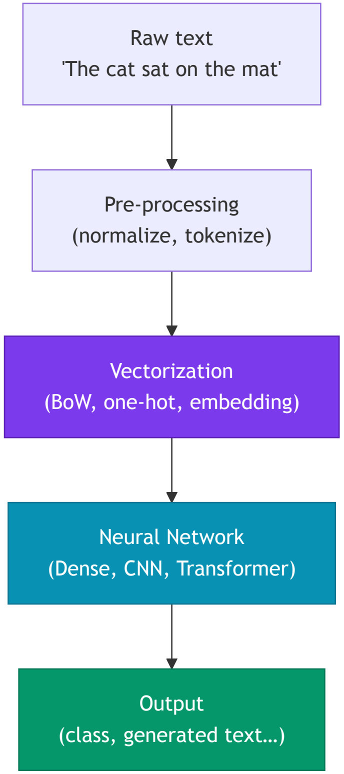
At its core, it's fancy regression

$$\hat{y} = f(\mathbf{x}, \theta)$$

- $\mathbf{x}$ = input text (token sequence)
- $\hat{y}$ = text, label, number, ...
- θ = neural network weights
- f = deep architecture (CNN, RNN, Transformer...)

Central question of this lecture:

How do we represent $\mathbf{x}$ (text) as a numerical vector so the network can learn?



Part 2 — Text Pre-processing

Pre-processing pipeline

Standardize

- Lowercase
- Remove punctuation/accents
- *Stop words* (sometimes)
- Stemming/lemmatization (sometimes)

Tokenize

- Split into tokens
- Default: whitespace
- Alternative: subwords (BPE)
- N-grams (bigram, trigram...)

Index

- Assign unique integer to each token
- Builds the **vocabulary** $\mathcal{V}$
- Special token <UNK> for out-of-vocabulary words

Encode

- Map integer → vector
- Simplest: one-hot
- Better: word embedding

Pre-processing example

Original text:

```
"Hola! What do you picture when you think of traveling to Mexico? Sipping a real margarita while soaking up the sun on a laid-back beach in Puerto Vallarta?"
```

After standardize (lowercase, remove punctuation, stop words, stemming):

```
`hola picture think travel mexico sip real margarita soak sun laidback beach puerto vallarta`
```

After tokenize (by whitespace):

```
`["hola", "picture", "think", "travel", "mexico", "sip", "real", "margarita", "soak", "sun", "laidback", "beach", "puerto", "vallarta"]`
```

After index (vocabulary with 50 k tokens):

```
`[4231, 1807, 2943, 8821, 9012, 3344, 771, 18432, 6621, 488, 22301, 993, 19034, 19035]`
```

One-hot encoding

Idea: each vocabulary token gets a vector with 1 at its position and 0 everywhere else.

```
Vocabulary (|V| = 5):  cat, dog, fish, bird, mouse

cat   → [1, 0, 0, 0, 0]
dog   → [0, 1, 0, 0, 0]
fish  → [0, 0, 1, 0, 0]
bird  → [0, 0, 0, 1, 0]
mouse → [0, 0, 0, 0, 1]
```

- Vector dimension = $|\mathcal{V}|$ (vocabulary size)
- Special token <UNK> (index 0) for out-of-vocabulary words
- **Sparse** representation — only one non-zero element

Problem: for $|\mathcal{V}| = 50{,}000$, each token becomes a 50 k-dimensional vector.

❌ **Very high dimensionality** → many parameters, overfitting

❌ **No semantic similarity** — distance between any two one-hot vectors is always $\sqrt{2}$, regardless of meaning

```python
from sklearn.preprocessing import LabelBinarizer

vocab = ["cat", "dog", "fish", "bird", "mouse"]
lb = LabelBinarizer()
lb.fit(vocab)
print(lb.transform(["cat", "fish"]))
# [[1 0 0 0 0]
#  [0 0 1 0 0]]
```

Part 3 — Bag of Words

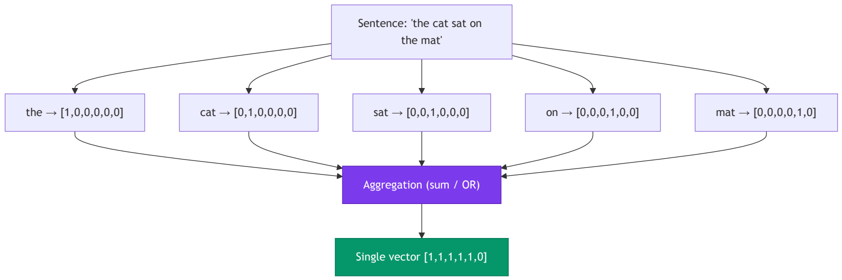
From per-token to per-document vectors

The problem: sentences have different lengths.

A 5-token sentence gives a $5 \times |\mathcal{V}|$ matrix.
A 20-token sentence gives $20 \times |\mathcal{V}|$.

Dense networks need **fixed-size input**.

Solution: aggregate (*pool*) all token one-hot vectors into a single vector of size $|\mathcal{V}|$.



Bag of Words — two variants

Count encoding (sum of one-hot vectors)

Counts how many times each token appears.

```
"the cat sat on the mat"
"the mat belonged to the cat"

vocab = [the, cat, sat, on, mat, belonged, to]

Sentence 1: [2, 1, 1, 1, 1, 0, 0]
Sentence 2: [2, 1, 0, 0, 1, 1, 1]
```

If "cat" appears 3×: that component = 3.

Multi-hot encoding (OR of one-hot vectors)

Indicates only whether the token is present (0 or 1).

```
"the cat cat cat sat"

Count:     [1, 3, 1, 0, 0, 0, 0]
Multi-hot: [1, 1, 1, 0, 0, 0, 0]
```

Multi-hot ignores frequency; useful when presence matters more than count.

Both are forms of **Bag of Words** — the name comes from treating text as a "bag" of words without order.

N-grams — capturing local context

Unigram: 1 token at a time (standard BoW)

Bigram: consecutive pairs of tokens

Trigram: consecutive triples of tokens

```
Sentence: "the cat sat on the mat"

Unigrams:  ["the", "cat", "sat", "on", "the", "mat"]

Bigrams:   ["the cat", "cat sat",
            "sat on", "on the", "the mat"]

Trigrams:  ["the cat sat",
            "cat sat on",
            "sat on the",
            "on the mat"]
```

Why use n-grams?

- ✅ Preserves some local context
- ✅ "not good" ≠ "good" (bigram captures negation)
- ✅ Simple, works well for text classification

- ❌ Vocabulary grows explosively: $|\mathcal{V}|^n$
- ❌ Still ignores long-range dependencies
- ❌ "good but not great" would need a 4-gram

```
from sklearn.feature_extraction.text import CountVectorize
cv = CountVectorizer(ngram_range=(1, 2))
X = cv.fit_transform(["the cat sat on the mat"])
```

Bag of Words limitations

❌ **Loses word order**

"The dog bit the man"

and

"The man bit the dog"

have the **same** BoW vector.

❌ **High dimensionality**

$|\mathcal{V}|$ can be 50 k–500 k.

Every input has that dimension regardless of sentence length.

Many parameters → overfitting, slow.

❌ **No semantics**

"movie" and "film" are equidistant from each other and from "banana".

BoW does not capture that they are related words.

Summary: BoW is a useful, simple baseline, but for tasks requiring semantic understanding or long-range dependencies we need something better — **Word Embeddings**.

Part 4 — Word Embeddings

The problem with one-hot representations

Problem 1: High dimensionality

Vocabulary of 50 k words → 50 k-dimensional vector.

Most elements are zero (**sparse**).

Problem 2: No semantic distance

```
"movie"  → [1, 0, 0, 0, 0, ...]
"film"   → [0, 1, 0, 0, 0, ...]
"banana" → [0, 0, 1, 0, 0, ...]

cosine("movie",  "film")   = 0
cosine("movie",  "banana") = 0
# All words are equally "distant"!
```

The ideal solution

Dense and **compact** vectors (dimension 50–300) where **geometric distance** reflects **semantic distance**.

```
"movie"  ≈ [0.2,  0.8, -0.3, ...]  # compact
"film"   ≈ [0.19, 0.79, -0.28, ...] # close to "movie"
"banana" ≈ [-0.6, 0.1,  0.9, ...]  # far from both

cosine("movie", "film")   ≈ 0.97  ✅
cosine("movie", "banana") ≈ 0.12  ✅
```

The intuition: "you shall know a word by the company it keeps"

"You shall know a word by the company it keeps."

— John Firth, linguist (1957)

Similar contexts → related words

```
"The acting in the ______ was superb."

→ movie      ✅
→ film       ✅
→ musical    ✅
→ truck      ❌
→ banana     ❌
```

Since "movie", "film" and "musical" appear in the **same contexts**, we infer they are semantically related.

Key insight

Count how often two tokens co-occur in the same contexts and learn vectors that **approximate** that co-occurrence matrix.

This is the foundation of two highly influential algorithms:

- **Word2Vec** (Mikolov et al., 2013) — local prediction
- **GloVe** (Pennington et al., 2014) — global co-occurrence

Word2Vec — two models

CBOW (*Continuous Bag of Words*)

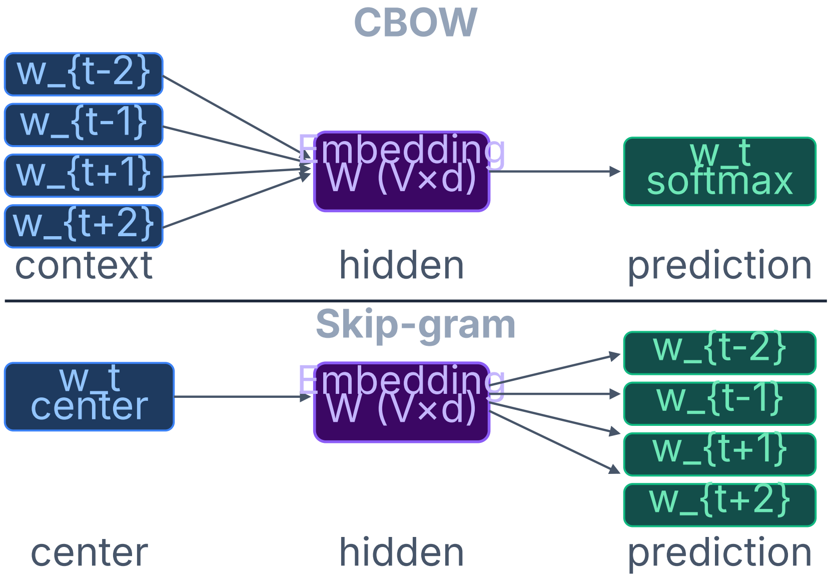
Given the context, predict the center word.

```
Context → Center word
["The", "acting", "was", "superb"]  →  "in the ___"
```

Skip-gram

Given the center word, predict the context words.

```
Center word → Context
"movie"  →  ["The", "acting", "was", "superb"]
```



Negative Sampling — avoids softmax over |
context) a real pair or random? → binary clas
5–20 negatives per step → O(k) instead of O(

Word2Vec — Negative Sampling

Problem: softmax over 50 k tokens at every step is costly — $O(|\mathcal{V}|)$.

Solution — Negative Sampling:

Instead of normalizing over the whole vocabulary, train a **binary classifier**:

$$P(D{=}1 \mid w, c) = \sigma(\mathbf{v}_c^\top \mathbf{v}_w)$$

"Did this (word, context) pair come from the real corpus?"

```
Positive pair: ("movie", "acting")  → label 1
Negative pairs (k random samples):
  ("movie", "table")    → label 0
  ("movie", "cloud")    → label 0
  ...
```

Objective with negative sampling:

$$\mathcal{L} = \log \sigma(\mathbf{v}_c^\top \mathbf{v}_w) + \sum_{i=1}^{k} \mathbb{E}_{w_i \sim P_n} \left[\log \sigma(-\mathbf{v}_{w_i}^\top \mathbf{v}_w)\right]$$

✅ Per-step complexity: $O(k)$ with $k \ll |\mathcal{V}|$ (typically $k = 5$–20)

✅ Empirically produces embeddings of similar quality to full softmax

More frequent words have a higher probability of being sampled as negatives (distribution $P_n(w) \propto f(w)^{3/4}$).

GloVe — Global Vectors

Intuition: Word2Vec uses local windows. GloVe uses **global co-occurrences** of the corpus.

Step 1: Count X_{ij} = how many times word j appears in the context of word i across the entire corpus.

```
             movie  film  banana  ...
movie    [  –,   8432,    12,   ... ]
film     [ 8432,  –,       9,   ... ]
banana   [   12,   9,     –,    ... ]
```

"movie" and "film" co-occur a lot; both rarely co-occur with "banana".

Step 2: Learn vectors that approximate the co-occurrence matrix.

Log-bilinear model:

$$\log X_{ij} = \mathbf{w}_i^\top \mathbf{w}_j + b_i + b_j$$

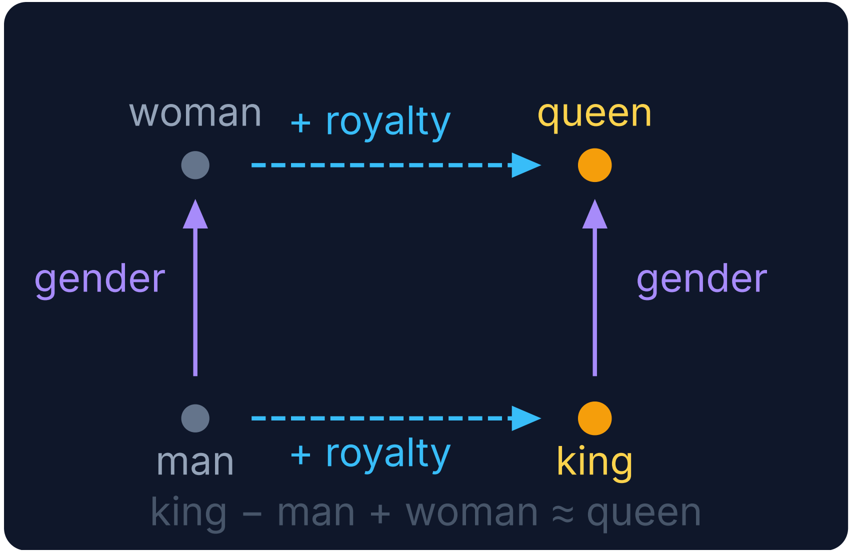
Objective (weighted least squares):

$$\mathcal{L} = \sum_{i,j} f(X_{ij})\left(\mathbf{w}_i^\top \mathbf{w}_j + b_i + b_j - \log X_{ij}\right)^2$$

where $f(X_{ij})$ is a weighting function that prevents very frequent pairs from dominating training.

At the end, biases b are discarded and only $\mathbf{w}$ is used.

Geometric properties of embeddings



2D projection — each dimension captures a latent concept that emerges from training.

Types of relations that emerge

Gender
king → queen · actor → actress · man → woman

Country → Capital
France → Paris · Brazil → Brasília · Japan → Tokyo

Superlative / Degree
good → better → best · big → bigger → biggest

Verb tense
run → ran · sing → sang · go → went

Antonyms
hot ↔ cold · fast ↔ slow · big ↔ small

⚠️ These patterns emerge from the corpus — and also reflect its **biases** (Wikipedia, Common Crawl...).

Static vs contextual embeddings

Static embeddings

(Word2Vec, GloVe, FastText)

- One fixed vector per word, independent of context
- Fast to compute, low memory footprint
- Good for simple classification tasks

❌ "bank" (financial) has the same vector as "bank" (riverbank)

Contextual embeddings

(ELMo, BERT, GPT)

- Vector depends on all other tokens in the sentence
- Captures polysemy and long-range dependencies
- Foundation of modern LLMs

✅ "bank" generates different vectors in each context — computed by a **Transformer** (Lecture 6!)

This lecture covers static embeddings. **Lecture 5** covers RNNs/LSTMs and **Lecture 6** covers Transformers and contextual embeddings.

Part 5 — Embeddings in Code

`nn.Embedding` — PyTorch

What it is: a lookup table — maps integer indices to dense vectors.

```
import torch
import torch.nn as nn

# vocab_size = vocabulary size
# embed_dim  = embedding dimension
embed = nn.Embedding(
    num_embeddings=10_000,  # |V|
    embedding_dim=128        # d
)

# Token indices for a batch of 2 sentences
# (zero-padded to max_len=5)
tokens = torch.tensor([
    [42, 871, 3, 0, 0],
    [7, 1024, 55, 810, 2]
])  # (batch=2, seq_len=5)

out = embed(tokens)  # (2, 5, 128)
print(out.shape)     # torch.Size([2, 5, 128])
```

Loading pre-trained weights (GloVe)

```
import numpy as np

# Assumes glove_matrix: (|V|, d) pre-loaded
embed = nn.Embedding(
    num_embeddings=vocab_size,
    embedding_dim=100
)
embed.weight = nn.Parameter(
    torch.tensor(glove_matrix, dtype=torch.float32)
)

# Optional: freeze weights during fine-tuning
embed.weight.requires_grad = False
```

`nn.Embedding` is equivalent to an `nn.Linear` layer without bias, but with O(1) lookup — it does not perform a full matrix multiplication over all sentences.

`Embedding` layer — Keras

Full pipeline with Keras

```
import tensorflow as tf
from tensorflow import keras

# 1. Vectorization (STI)
vectorizer = keras.layers.TextVectorization(
    max_tokens=10_000,    # |V|
    output_mode='int',    # returns indices
    output_sequence_length=64   # max_len
)
vectorizer.adapt(train_texts)  # learns vocabulary

# 2. Embedding
embed = keras.layers.Embedding(
    input_dim=10_000,     # |V|
    output_dim=128,       # d
    mask_zero=True        # ignore padding
)
```

Full model

```
inp   = keras.Input(shape=(1,), dtype='string')
x     = vectorizer(inp)          # (batch, 64)
x     = embed(x)                 # (batch, 64, 128)
x     = keras.layers.GlobalAveragePooling1D()(x)
      # mean over seq_len → (batch, 128)
x     = keras.layers.Dense(64, activation='relu')(x)
out   = keras.layers.Dense(num_classes,
                           activation='softmax')(x)

model = keras.Model(inp, out)
model.compile(
    loss='sparse_categorical_crossentropy',
    optimizer='adam',
    metrics=['accuracy']
)
```

Global Average Pooling — why it works

After `Embedding`, each sentence is a **matrix** $(T \times d)$. We need a **fixed vector** for the dense layer.

```
"the cat sat"  →  Embedding
[[0.2, -0.1, 0.8, ...],  ← "the"
 [0.7,  0.3, 0.1, ...],  ← "cat"
 [0.4, -0.2, 0.6, ...]]  ← "sat"   shape: (3,d)

GAP → mean over T
→ [0.43, 0.0, 0.5, ...]            shape: (d,)
```

Alternatives to GlobalAveragePooling

Flatten — concatenates all vectors into $(T \times d)$, requires fixed length.

GlobalMaxPooling1D — maximum per dimension, preserves strongest activations.

[CLS] token / attention — Transformer strategies (Lecture 6). BERT uses `[CLS]`.

GAP is equivalent to a BoW weighted by embeddings — simple yet surprisingly effective.

Part 6 — Application: Text Classification

Musical genre classification

Task: given a song lyric excerpt, classify it as hip-hop, pop or rock.

> *"I grew up on the crime side, the New York Times side… Had secondhands, Mom's bounced on old man…"* → **hip-hop** 🎤

> *"I walked through the door with you But something about it felt like home somehow…"* → **pop** 🎵

BoW + Embedding architecture

Full code — PyTorch

```python
import torch, torch.nn as nn, torch.optim as optim

class TextClassifier(nn.Module):
    def __init__(self, vocab_size, embed_dim, num_classes):
        super().__init__()
        self.embed   = nn.Embedding(vocab_size, embed_dim, padding_idx=0)
        self.dropout = nn.Dropout(0.3)
        self.fc1     = nn.Linear(embed_dim, 64)
        self.fc2     = nn.Linear(64, num_classes)

    def forward(self, x):          # x: (batch, seq_len)
        x = self.embed(x)          # (batch, seq_len, embed_dim)
        x = x.mean(dim=1)          # GAP: (batch, embed_dim)
        x = self.dropout(x)
        x = torch.relu(self.fc1(x))
        return self.fc2(x)         # (batch, num_classes)

model     = TextClassifier(vocab_size=10_000, embed_dim=64, num_classes=3)
criterion = nn.CrossEntropyLoss()
optimizer = optim.Adam(model.parameters(), lr=1e-3)

for epoch in range(10):
    for tokens, labels in train_loader:
        logits = model(tokens)
```

Comparison: BoW vs Embeddings

	BoW (count/multi-hot)	Embedding + GAP
Representation	Sparse ($\|V\|$-dimensional)	Dense (d-dim, $d \ll \|V\|$)
Semantics	❌ No semantic distance	✅ Similar words are close
Word order	❌ Ignored	❌ Ignored (with GAP)
Parameters	None (engineered)	$\|V\| \times d$ (learned)
Transfer	Only with external embeddings	✅ Pre-train on large corpus
Cost	Low	Moderate

To capture word order and long-range dependencies → sequential architectures: RNNs and LSTMs **(Lecture 5)**, and ultimately Transformers **(Lecture 6)**.

Summary

Pre-processing

- S→T→I→E pipeline
- Standardization, tokenization, vocabulary
- One-hot encoding: sparse, no semantics

Bag of Words

- Count and multi-hot encoding
- N-grams for local context
- Limitations: order lost, high dimensionality

Word Embeddings

- Word2Vec (skip-gram, negative sampling)
- GloVe (global co-occurrence, least squares)
- Dense vectors with semantic distance

In code

- `nn.Embedding` (PyTorch)
- `Embedding` + `TextVectorization` (Keras)
- GlobalAveragePooling1D as weighted BoW

Remaining limitations

- No order with GAP
- Static embeddings: polysemy
- → We need Transformers!

Lecture 5 — RNNs and LSTMs

- Hidden state and sequential processing
- BPTT and the vanishing gradient problem
- LSTMs and GRUs as solutions

Next lecture

Lecture 5 — RNNs and LSTMs

- Hidden state and sequential processing
- BPTT and the vanishing gradient problem
- LSTM: memory cells and gates
- GRU: more efficient variant
- Applications: classification and seq2seq

For this week

- Notebook `lec04-embeddings.ipynb`:
  - Pre-processing pipeline with Keras/PyTorch
  - Train embeddings from scratch vs pre-trained GloVe
  - Musical genre classification
  - Embedding visualization with t-SNE

Thank you! Questions?

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Adapted from MIT 15.773 Hands-on Deep Learning (Farias & Ramakrishnan, 2024)